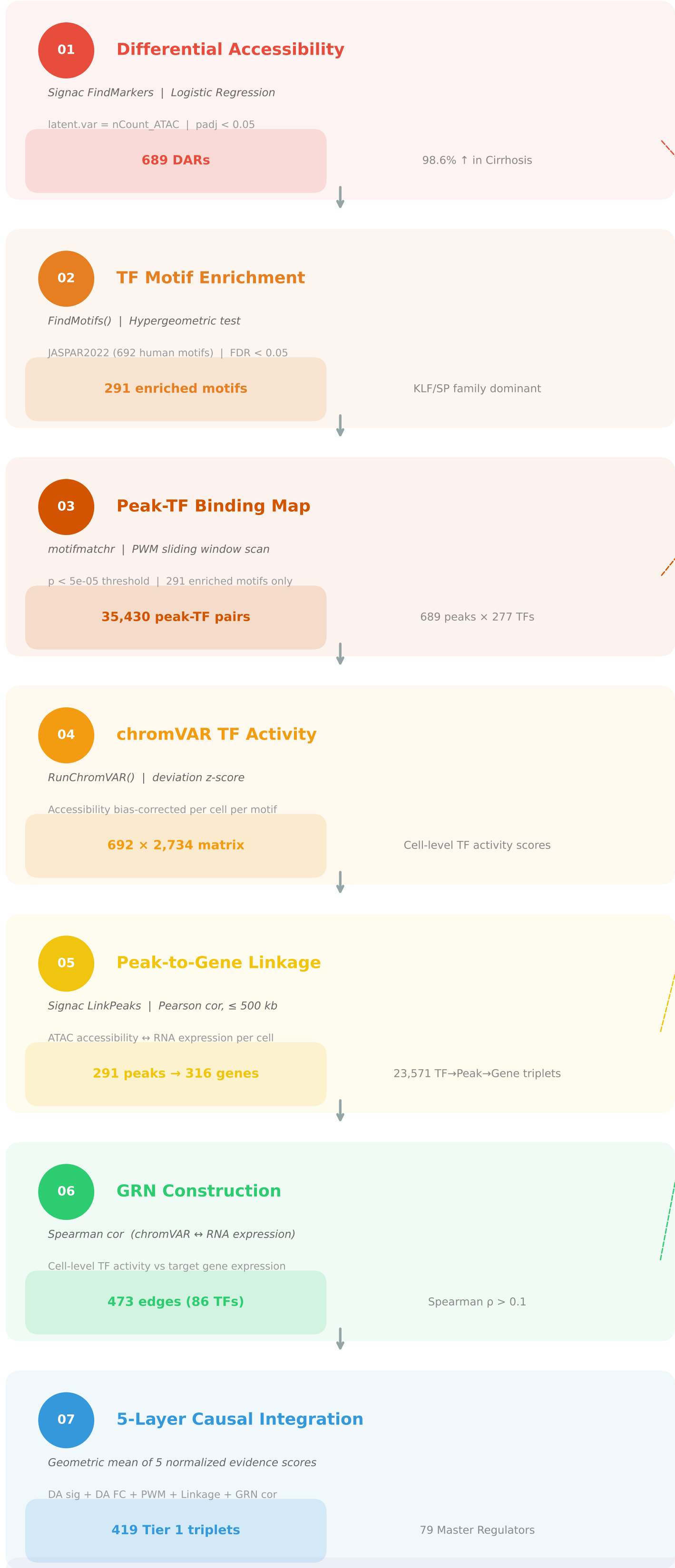


ATAC-seq Regulatory Analysis Pipeline

HCC Multiome | TC Hepatocytes | Cirrhosis vs Hepatitis



Causal Evidence Chain

TF → DAR → Gene (3-node regulatory path)



5 Evidence Layers

DA Significance	$-\log_{10}(\text{padj}) / 30$	Step 01
DA Effect Size	$ \log_2\text{FC} / 5$	Step 01
Motif PWM Score	PWM score / 20	Step 03
Peak-Gene Linkage	$ z\text{-score} / 10$	Step 05
TF-Gene Correlation	$ \text{Spearman } \rho / 0.4$	Step 06

Composite = $(ev_1 \times ev_2 \times ev_3 \times ev_4 \times ev_5)^{1/5}$

Evidence Tier Classification

Tier 1	All 5 evidence layers > 0	419	(1.8%)
Tier 2	PWM + Linkage + DA > 0	17,200	(73.0%)
Tier 3	DA + Linkage > 0	5,952	(25.2%)

Top 10 Tier 1 Master Regulators

Rank	TF	Targets	DARs	Score
1	HNF4A	55	43	0.301
2	NR1H2::RXRA	31	18	0.248
3	RXRG	28	23	0.229
4	NR2F1	22	19	0.260
5	RXRB	22	16	0.221
6	HNF4G	21	17	0.321
7	CTCF	19	17	0.306
8	HNF1B	16	18	0.277
9	TEAD4	13	10	0.255
10	HNF1A	11	14	0.251

Key Finding

HNF4A

Hepatocyte Nuclear Factor 4 Alpha

#1 Master Regulator of chromatin remodeling

in Cirrhosis-associated Hepatocytes

Notable HNF4A Target Genes

CYP3A4	Drug metabolism (cytochrome P450)
AGT	Angiotensinogen — blood pressure
GPT	ALT enzyme — liver function marker
SEC14L2	Lipid transport & cholesterol
TF	Transferrin — iron transport
POR	P450 oxidoreductase
SLC27A5	Bile acid metabolism

55

target genes

43

mediating DARs

0.301

causal score